

claude-windows / 45ea3a91-654d-4972-9cd3-65f35db54caf

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\45ea3a91-654d-4972-9cd3-65f35db54caf\subagents\workflows\wf_2a64bc88-2a8\agent-a2b1c80b0d2e7fa8b.jsonl`  
- SHA-256: `604e2b6d0e76fbd7f0c66c7a3017c70fa6a0775ca496608e4a11d7e324476e9d`  
- Source modified: `2026-06-10T05:43:16+00:00`  
- Imported at: `2026-07-05T16:48:27+00:00`  
- project: `wf_2a64bc88-2a8`  
- session_id: `45ea3a91-654d-4972-9cd3-65f35db54caf`
```

Transcript

1. user (2026-06-10T05:41:39.832Z)

CONTEXT ? two sibling repos on a Windows machine:

- INDEXER: C:/Users/avidu/Projects/badminton-highlight-indexer ? local AI badminton(->racket-sports) highlight indexer + rally detector. FastAPI + SQLite + ffmpeg + GPU CV via WSL; Gemini = paid cloud AI. docs/ tree is rich. Current branch docs/next-steps-multisport (master = main branch).

- COLLECTOR: C:/Users/avidu/Projects/sports-data-collector ? golden-label acquisition/curation/ingestion CLI (system-of-record candidate for training corpus).

Project state (June 2026): scaffolding complete; owned-model roadmap agreed (docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md) but NO platform/owned-model implementation started; next committed platform slice = async job model. Licensing guardrail: MIT/Apache-2.0/BSD only for code+data+weights; paid LLMs are inference-only, never training-data sources. ENVIRONMENT RULES: this dev environment has NO GPU/torch/cv2 ? do NOT attempt to run the video pipeline or import torch/cv2 modules. Test suites are offline/fully-mocked and SAFE to run. You are READ-ONLY: never edit/create/delete repo files; running tests and git read commands is allowed.

QUALITY BAR: cite file paths (and line numbers where possible) for every finding. Report what IS in the code/docs, not what you assume. Be skeptical and concrete. Your final structured output is consumed programmatically by an orchestrator ? return raw data, not prose for a human.

ROLE: adversarial verifier. An auditor ("docs-strategy") claimed the finding below. Try to REFUTE it against the actual files. Default to isReal=false if the evidence does not hold up, the behavior is documented/deliberate (e.g. listed in docs/CODE_MAP.md gotchas or DEFERRED_HARDENING_PLAN.md as a known deferral ? unless the auditor adds genuinely NEW consequence), or the severity is materially overstated.

FINDING: {"title":"Gemma-4 stance contradiction: approved base (CLAUDE.md/PLATFORM/COMMERCIALIZATION) vs AMBER/bench-only (plan/study banner)","severity":"high","area":"owned-model licensing","file":"C:/Users/avidu/Projects/badminton-highlight-indexer/CLAUDE.md:34","evidence":"CLAUDE.md:34: 'Owned-model base: Apache-2.0, video-capable (Qwen-VL / **Gemma 4**)' ? presented as an approved option. PLATFORM_ARCHITECTURE.md:304: 'Gemma 4 multimodal (Apache-2.0, 2026) is a clean Google-stack alternative to evaluate'; COMMERCIALIZATION.md C14 (line 43) lists '...or Gemma 4' among clean bases; checkpoint README:25 same. But OWNED_MODEL_IMPLEMENTATION_PLAN.md:42-43: '**Gemma-4 stays AMBER and bench-only** ? its Apache grant is likely but its Prohibited-Use posture is unconfirmed (counsel needed), and its video/timestamp capability is disputed; not the primary extractor.' OWNED_MODEL_TRAINING_STUDY.md:22-23 banner: 'Gemma 4 stays AMBER and bench-only (NOT \"confirmed clean\" as the table below says)'. The live docs were never downgraded to AMBER.\"","why_it_matters":"A reader of CLAUDE.md or PLATFORM could select Gemma 4 as a fine-tune base ? exactly the action the newest owner-reviewed doc says requires counsel first. Given the project rejected Gemma 3 for a viral license, an unvetted Gemma-4 adoption is a real legal exposure."}

Steps: open the cited file(s) and any counter-evidence files; verify quotes/line claims; judge severity honestly. Cheap greps allowed; do NOT run the pipeline. Return isReal, confidence, and a 1-3 sentence verdict (with corrected detail if partially right).

2. user (2026-06-10T05:41:46.156Z)

```
1      # CLAUDE.md ? project memory / agent handoff
2
3      Local, AI-powered badminton (? racket-sports) highlight indexer + rally detector**:
4      FastAPI + SQLite + ffmpeg + GPU CV/AI, with paid Gemini as the cloud AI. Sibling
5      repos: `sports-data-collector`, a VLC-based `rally-annotator`.
6
7      ## Read these first
8      - docs/README.md ? doc index. docs/CODE_MAP.md ? code navigation + data
9        contracts (the cold-start map; read before touching code).
10     - Strategy (post-scaffolding): docs/COMMERCIALIZATION.md,
11       docs/PMF_MARKET_RESEARCH.md,
12       docs/PLATFORM_ARCHITECTURE.md, docs/DEFERRED_HARDENING_PLAN.md. Also
13       docs/ROADMAP.md
14       (offline-segmenter narrative) and docs/BACKLOG.md.
15     - History: docs/archives/ ? ADRs (decisions/DECISIONS.md), research, and
16       checkpoints/ (latest: 2026-06-strategy-foundation/). Archive policy: only move
17       terminal-state (DONE/REJECTED) work; live docs stay at docs/ top level.
18
19     ## Current state (June 2026)
20     - Scaffolding complete; the post-scaffolding strategy foundation is laid and
21       merged
22       (#52/#53/#55/#57/#58). No platform/owned-model implementation has started.
23     - Next slice = async job model (DEFERRED_HARDENING_PLAN.md #16), single-tenant.
24     - Do not start implementation without explicit owner approval.
25
26     ## Architecture in one breath
27     - Pluggable registries (ABC + Registry singleton): segmenters / providers /
28       validators
29       / sports / annotations ? see backend/pipeline/segmenters/base.py. Future:
30       StorageBackend + ComputeBackend registries (PLATFORM $3).
31     - video_id = MD5 of the file ? backend/utils/hashing.py::compute_video_id (one
32       helper).
33     - Typed config = backend/config/ (Pydantic). DB schema evolves via the PRAGMA
34       user_version migration ladder in backend/infrastructure/database.py.
35
36     ## Committed guardrails (non-negotiable)
37     - Paid LLMs (Gemini/OpenAI/Anthropic) = inference / serving / fallback ONLY ? never a
38       training data source (terms/copyright). Training teachers/bases = open Apache-2.0
39       only.
40     - Owned-model base: Apache-2.0, video-capable (Qwen-VL / Gemma 4); reject Gemma
41       (viral license). Exclude minors from the training pool; per-user training data
42       needs a
43       separate explicit opt-in.
44     - v1 = rally + highlight + history + correction loop; gait/biometrics strictly
45       future
46       (GDPR Art. 9).
47     - Generic data schema; coach ? athlete is an optional overlay, not baked in.
48
49     ## Workflow conventions
50     - Branch off master; ONE PR per work item; NO PR stacking. Don't open a PR unless
51       asked.
52     - Tests: python run_all_tests.py (offline, fully mocked). Python 3.8+; ffmpeg
53       required.
54     - Never push to the default branch directly; descriptive commit messages.
55     - ?? This dev container lacks GPU / torch / cv2 / numpy ? the full pipeline can't run
56       here.
57     - Verify pure-logic paths and hand machine-side checks (real video, GPU/WSL suite) to
58       the owner.
```

3. user (2026-06-10T05:41:46.891Z)

```

1      # Owned-model implementation plan (roadmap) ? DRAFT for owner review
2
3      **Status:** PLAN / roadmap. No owned-model *implementation* starts until the owner
greenlights a specific
4      milestone. Paid Gemini stays the always-ready inference/fallback throughout (never a
training source).
5      **Grounded in:** `docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md`, the n=6 LOVO +
learned prototype
6      (`docs/archives/quality-iterations/2026-06-multivideo-lovo/`), and a verified strategy
workflow (2026-06-09).
7
8      ## Design principle (owner directive): *envision now, build later*
9      Every deferred piece (conversational-QA, on-device, multisport, the VLM) must be
**architecturally envisioned
10     now** so we never hit a mammoth refactor. We **design the seams** (data contracts,
export boundaries, the
11     event store) up front and **implement behind them** when each milestone is greenlit.
Punting a *feature* is fine;
12     punting its *seam* is not.
13
14     ## North Star (reframed per PR #61 review)
15     - **Immediate deliverable:** a **training framework + harness** that produces a model
with **very high precision/
16     recall for extracting highlights + rallies from a clip**. *Quality is the first-order
objective.*
17     - **Eventual product:** a **single, sport-agnostic, conversational** video model ?
upload a clip, then ask
18     contextual questions ("longest rally", "make a clip of all service faults", "rallies
over 20 shots").
19     - **On-device is LATER** (a model-compression goal so power users save upload
cost/hassle) ? definitely on the
20     roadmap, but **not** the immediate target. Don't over-index the strategy on it now.
21     - **The moat = the consented dataset + the eval harness, not the weights.**
22
23     ## Hard guardrails (non-negotiable ? "never corrupted")
24     - **Commercial-clean licensing for EVERY dependency ? code, data, AND model weights: MIT
/ Apache-2.0 / BSD only.**
25     (Ratifies the prior "Apache-2.0 only" to include MIT/BSD.) Reject viral/NC/custom-
restrictive: Gemma 3 (viral),
26     **Gemma 3n** (Gemma Terms follow derivatives ? even cleaner reject), Llama 4 (700M-MAU
cap), Commons-Clause repos.
27     - **Never train on paid-LLM (Gemini/OpenAI/Anthropic) outputs.** Also forbidden: public
grounding-CoT datasets that
28     were Gemini-generated (e.g. TVG-R1's 56K) ? we must originate our own CoT supervision
(a real, binding cost).
29     - **Exclude minors; per-user training data needs a separate explicit opt-in; split data
by match.**
30     - ? **Base-model pretraining provenance is unauditabile for *every* open VLM (incl.
Qwen3-VL).** Our "no paid-LLM
31     training" rule is enforceable at *our fine-tuning data* layer, not the base's
pretraining ? an **explicit owner
32     risk-acceptance** is required to use any open VLM. (Governed by the base's license,
not our pipeline.)
33
34     ## The model strategy (verified 2026-06-09)
35     **Framework ? base** (the Gemma-4-vs-ms-swift clarification): a *framework* is the forge
(runs SFT/GRPO, emits
36     weights); a *base* is the metal you forge. You pick one framework and feed it any clean
base.
37     - **Framework: ms-swift (Apache-2.0)** ? the one forge that does native **video + audio
+ timestamp-grounding +
38     GRPO/RFT with a custom video reward** in one stack. Keep **unsloth** (Apache-2.0 core;
*avoid its AGPL Studio UI*)
39     as the 8 GB-local SFT/compression tool for the deferred on-device goal. ? Point-in-
time snapshot ? re-verify at impl.

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40     - **Base (when we reach the VLM): Qwen3-VL (Apache-2.0, uniform across sizes)** ? long-
video context, purpose-built
41     **timestamp emission** (second-level localization), multi-turn video chat. Re-confirm
the exact size's LICENSE
42     pre-release. **Gemma-4 stays AMBER and bench-only** ? its Apache grant is likely but
its Prohibited-Use posture is
43     unconfirmed (counsel needed), *and* its video/timestamp capability is disputed; not
the primary extractor.
44     **InternVL3 (MIT/Apache)** = clean fallback if a Qwen blocker appears. ? base swap is
cheap *only within
45     timestamp-capable Apache bases* (timestamp data-format + grounding-CoT are base-
specific).
46     - **START WITH THE TAL HEAD, NOT THE VLM** (the decisive call): fine-tuned VLMs **miss
strict temporal boundaries**
47     (best RL-post-trained video VLM ? R@0.7 50% on open-domain benchmarks; "coarse
regions, not accurate boundaries").
48     **Boundary accuracy IS the product.** The head (`rally_seq_proto.py`) is already de-
risked on our corpus (held-out
49     AUC 0.83?0.92), trains in minutes at ~4.5 GB on the 2070 Super ($0), is ~1M params
(trivially compressible later),
50     and **doubles as a pseudo-label teacher + the honest baseline the VLM must beat.** So:
**head now ? VLM later**,
51     strictly sequenced. When the VLM comes, prefer **two-stage** (VLM coarse-proposes,
head refines boundaries) over
52     wholesale replacement. *(Open: the VLM-ceiling number is open-domain; our amateur
footage may differ ? re-measure.)*
53
54     ## Conversational video-QA ? build the bookkeeping NOW, punt the model
55     **Punt the conversational *feature*** until extraction quality is solid (current LOVO
~0.583 ? "very high"; a chat UI
56     over a weak extractor exposes the weakness). **Reject end-to-end-VLM-answers-each-turn**
(expensive; bad at exact
57     counting; the strong teachers are license-forbidden). **But build the *seam* now** (the
brief requires it):
58     - **Architecture = "extract once, query many":** EXTRACTION (TAL head ? later VLM) emits
structured timestamped
59     spans ? a durable **per-video EVENT STORE** ? a **structured QUERY layer** (text-to-
SQL + ffmpeg clip-cut). The
60     cited queries ("longest rally", "all service faults", "rallies > 20 shots") are
**structured-numeric ? SQL, not a
61     VLM job** (SQL counts exactly where VLMs hallucinate).
62     - **Designed-now seam:** make the **per-video event-index schema a first-class data
contract** alongside the M0.1
63     corpus spine: per rally `{video_id(MD5), rally_ordinal, sport, start, end, shot_count,
ending_reason/fault_tags,
64     derived_clip_ref, provenance, confidence`. **Reserve the event/fault-tag taxonomy
(e.g. `service_fault`) up front**
65     even if unpopulated. ? This is a real DB delta (a `PRAGMA user_version` migration):
`play_segments` today has
66     `video_id/start_time/end_time/duration/server/receiver/metadata` only; `highlights`
isn't a table yet ? so it's an
67     *extension with new columns + a highlights table*, not a no-op. **Punt** the NL
parser/agent until quality clears the bar.
68     - Paid Gemini may answer genuinely open-ended edge questions **at inference** over our
structured index (allowed ?
69     it reasons over our data, it is not trained on its outputs).
70
71     ## Multisport (single sport-agnostic model)
72     - **Model/framework: UNCHANGED.** One shared TAL head + one Qwen3-VL base, fine-tuned
across sports (sport = an input
73     feature / per-sport calibration). The collector + indexer are already sport-generic. ?
**ACTION: rename the repo off
74     "badminton" ASAP** (owner emphatic) ? consistent with this.
75     - **What multisport changes = DATA + DETECTORS (scale up, the accepted "good
problem"): ** each new sport needs its own

```

76 labeled matches (split by match) + a **clean trajectory detector** to feed the head's features. **The binding blocker**
77 is per-sport detectors: **WASB** (badminton) has the unresolved **C3** checkpoint-provenance issue, and **clean MIT/**
78 Apache/BSD shuttle/ball detectors for tennis/TT/pickleball/padel are not yet identified. **Sequence by public-data**
79 richness: **badminton ? tennis/TT ? pickleball/padel.**
80 - **? Single-model-multisport-temporal-generalization is an OPEN HYPOTHESIS**, not proven ? our corpus is badminton-only,

4. user (2026-06-10T05:41:57.622Z)

Base model & training (Apache-2.0 only ? licensing is decisive)

Recommended near-term base: **Qwen2.5-VL-7B-Instruct** ? Apache-2.0, native long-video (dynamic-FPS sampling, temporal mRoPE), mature LoRA tooling. Upgrade path **Qwen3-VL-4B/8B** (Apache-2.0); **Gemma 4** multimodal (Apache-2.0, 2026) is a clean Google-stack alternative to evaluate.

- **Lock to Apache-2.0-clean bases** so you can fine-tune, **keep weights private**, gate behind subscriptions, and never owe open-sourcing or branding. (Note **Qwen2.5-VL** is Apache **only at 7B/32B** ? 3B/72B use the custom Qwen license.)

C13	**Beachhead validation**	**P0**	?	Market research (**PMF_MARKET_RESEARCH.md**) recommends **India badminton academies** as the #1 beachhead. Unvalidated.	Cheapest tests: 10?15 academy/coach discovery interviews; rally-detection P/R benchmark on 100+ real amateur clips; fake-door pricing (B2C ~\$99/yr, B2B academy ~\$600/yr). See §6.
C8	**Golden-set vendoring spend**	**P1**	?	ADR-006: **Roora** (Bengaluru) selected for an initial paid pilot; licensed/owned video only (no user uploads to vendors).	Confirm quality-per-dollar from the pilot; then GS-3 **HumanVendorProvider**. See **GOLDEN_SET_VENDORING.md**.
C3	**Shuttle-tracking weights provenance**	**P1**	? ?	**WASB-SBDT** **code is MIT** (clear); distributed **checkpoints are academic-data-trained** ? not covered by the MIT code grant. **MonoTrack/YOLO-NAS** banned (noncommercial).	Train-own weights (Phase-4 roadmap) **or** confirm per-checkpoint terms ? before commercial deploy.
C14	**Owned-model base licensing + training-data policy**	**P1**	? ?	Owned video-QA model (PLATFORM §5A) must build on **Apache-2.0-clean, video-capable** bases ? **Qwen2.5-VL-7B/32B** (Apache; 3B/72B are *not*), **Qwen3-VL-4B/8B**, or **Gemma 4** ? to keep weights private, gate, and monetize. **Reject Gemma 3** (viral derivative clause ? permanent Google PUP), Llama Vision (naming + 700M-MAU), **gpt-oss** (text-only). **Hard rule: never train on paid-LLM (Gemini/OpenAI/Anthropic) outputs** (competing-model clause). The moat is the **consented dataset + eval harness**, not the model.	Lock to Apache-2.0; **open-model teachers only**; separate explicit **training opt-in**; **exclude minors** from the training pool; per-clip consent ledger. See PLATFORM §5A/§7.
C6	**Inbound HEVC decode**	**P2**	? ?	Decoding users' GoPro H.264/**HEVC** uploads is unavoidable; HEVC pools are fragmented + highest-uncertainty.	Low/uncertain residual; confirm decode-side exposure with counsel before launch.
C7	**Hosting unit-economics**	**P2**	?	Per-video cost is *cheap* for Gemini (~\$0.05 input for a 10-min clip); risk is volume × latency, not unit price. Self-host/offline costs unmeasured.	**Benchmark \$/video & latency on the target VM** for A/B/C before pricing. See §5.
C10	**Value layer (product surface)**	**P2**	?	Minimal exports shipped (EDL/JSON/CSV + **basic_stats**, PR #51).	Deepen: per-rally tags/notes that feed golden, per-player/match stats, clip gallery. See §6.

is an **open hypothesis** (corpus is badminton-only; the once-cited RacketVision evidence was **refuted** ? it's

ball-tracking, not temporal).

- **Framework = ms-swift** (Apache-2.0) (only forge with native video+audio+timestamp-grounding+video-GRPO).

Base = Qwen3-VL (Apache-2.0, uniform sizes, timestamp emission) ? primary, **replacing the Qwen2.5-VL-7B framing**.

- **Gemma 4 stays AMBER and bench-only** (NOT "confirmed clean" as the table below says): Apache grant likely but

Prohibited-Use posture unconfirmed (counsel) + video/timestamp capability disputed ? not the primary extractor.

- **Start with the TAL head, not the VLM** (VLMs miss strict boundaries ? boundary accuracy is

the product). Head now ? VLM later.

- ****Immediate goal = a high-P/R extraction *harness*, not on-device**** (on-device deferred).
****Reframe supersedes the on-device-now framing.****

Thesis

See git history around #52/#53/#55/#57/#58 for exact snapshots.

Committed decisions / non-negotiables recorded here

- ****Paid frontier models (Gemini/OpenAI/Anthropic): inference / serving / fallback only ? never a training data source**** (competing-model terms). Teachers/bases = open Apache-2.0 only.
- ****Owned-model base = Apache-2.0-clean, video-capable**** (Qwen2.5-VL-7B/32B, Qwen3-VL, Gemma 4);
****reject Gemma 3**** (viral license clause); ****minors excluded**** from the training pool; per-user training needs a separate opt-in.
- ****v1 = rally detection + highlight compilation + per-user history + coach/user correction loop**** ("assistant-coach lite"); ****gait/biomechanics strictly future**** (Art. 9).
- ****Generic data schema**** (User ? VideoCollection ? Video ? Highlight ? HighlightMetadata); coach ? athlete is an ****optional overlay**** (a market-fit hypothesis).

What comes next (per owner)

- First implementation slice = ****async job model (DEFERRED #16)****, single-tenant ? ****pending explicit go-ahead**** (no implementation started without approval).

5. user (2026-06-10T05:42:13.890Z)

C:\Users\avidu\Projects\badminton-highlight-indexer\CLAUDE.md:34:- ****Owned-model base:****
Apache-2.0, video-capable (Qwen-VL / Gemma 4); ****reject Gemma 3****
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\COMMERCIALIZATION.md:43:[Omitted long matching line]
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\NEXT_STEPS.md:11:rally-vs-non-rally problem). ****Framework = ms-swift, base = Qwen3-VL**** (Gemma-4 AMBER/bench) come ****later**** for the
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md:35:****Framework ? base**** (the Gemma-4-vs-ms-swift clarification): a ****framework**** is the forge (runs SFT/GRPO, emits
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md:42:pre-release. ****Gemma-4 stays AMBER and bench-only**** ? its Apache grant is likely but its Prohibited-Use posture is
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PLATFORM_ARCHITECTURE.md:269:> are open, Apache-2.0 models only** (Qwen-VL / InternVL / Gemma 4). Detail in the
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PLATFORM_ARCHITECTURE.md:304:****Qwen3-VL-4B/8B (Apache-2.0)****; ****Gemma 4 multimodal (Apache-2.0, 2026)**** is a clean
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PLATFORM_ARCHITECTURE.md:311:permanently bound to Google's ****updatable* Prohibited-Use Policy****; ****Gemma 4's switch to**
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PLATFORM_ARCHITECTURE.md:419:InternVL / Gemma 4 ? Apache-clean); **** (2) **** keep the paid API strictly as a ****runtime**
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PLATFORM_ARCHITECTURE.md:563:[Omitted long matching line]
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\archives\checkpoints\2026-06-strategy-foundation\README.md:25:- ****Owned-model base = Apache-2.0-clean, video-capable****
(Qwen2.5-VL-7B/32B, Qwen3-VL, Gemma 4); ****reject Gemma 3**** (viral license clause); ****minors excluded**** from the training pool; per-user training needs a separate opt-in.
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md:22:- ? ****Gemma 4 stays AMBER and bench-only**** (NOT "confirmed clean" as the table below says): Apache grant likely but
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md:37:| ****Gen-2 / Option B**** | 12?24+ mo, after scaffolding + healthy corpus | end-to-end multimodal VLM:
(frames+audio+pose+trajectory) ? rally [start,end] (+semantics); SFT cold-start then ****GRPO/RFT with a temporal-IoU-vs-golden reward**** | ****Qwen3-VL-4B-Instruct**** (Apache-2.0, on-device) primary; ****Qwen2.5-VL-7B**** fallback; ****Gemma 4 E4B/12B-Unified**** (Apache-2.0 + native

video+audio) end-state | rented 24?48 GB A100/L40S (does NOT fit 8 GB box) |
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md:98: Gemma 4 E2B/E4B/12B-Unified (Apache-2.0 + native video/audio), InternVL3 ?38B (MIT/Apache), ASFormer/
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md:102: (Commons Clause ? use `sj-li/MS-TCN2` MIT). ? **Gemma 4 size names = E2B/E4B/26B-A4B/31B/12B-Unified ? there is no plain "4B".**
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md:127:- **Gemma 4 "clean + native-video" rests on post-cutoff secondary sources** ? keep AMBER until a raw HF LICENSE
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md:147:3. Working Gemma-4 timestamp-emission recipe + exact checkpoint (E4B vs 12B-Unified)?

6. assistant (2026-06-10T05:42:21.985Z)

The newest live doc NEXT_STEPS.md:11 already reflects "Gemma-4 AMBER/bench", but other live docs don't. Let me check the PLATFORM context around lines 308-315 and 560-565.

7. user (2026-06-10T05:42:25.337Z)

> full. **Paid frontier models (Gemini / OpenAI / Anthropic) are used for *inference /
> serving / fallback* only ? never as a *training* data source.** Harvesting their outputs
> to train our owned model would breach their "no competing model" terms (a terms/copyright
> violation), so it is prohibited in both engineering and legal process. **Training teachers
> are open, Apache-2.0 models only** (Qwen-VL / InternVL / Gemma 4). Detail in the
> distillation constraint below.

Core principle: AI is a pluggable provider with a paid fallback

is Apache **only at 7B/32B** ? 3B/72B use the custom Qwen license.)
- ? **Reject for the flagship:** **Gemma 3** ? custom terms with a **viral derivative clause** (anything fine-tuned on it, *or even trained on its synthetic outputs*, stays permanently bound to Google's *updatable* Prohibited-Use Policy); **Gemma 4's switch to Apache-2.0 removes this**. **Llama 3.2 Vision** ? custom license forces a "Llama-?" model name + "Built with Llama" badge and a 700M-MAU ceiling. **`gpt-oss`** ? Apache-2.0 but **text-only**, so usable only as an optional text-reasoning layer, never the video encoder.
- **Training method:** **QLoRA** (? r=64, ?=128) on the 7B ? fits one prosumer GPU and video-QA assistant is squarely a "competing model": **Gemini** ("may not use the Services to develop models that compete"), **OpenAI** ("may not use Output to develop models that compete"), **Anthropic** (bars using Outputs to train competing models). Two safe lanes: **(1)** generate synthetic QA with **open-model teachers only** (Qwen-VL / InternVL / Gemma 4 ? Apache-clean); (2) keep the paid API strictly as a **runtime fallback** ? serving answers is fine, *harvesting its outputs to train the model that replaces it is not*. Write a hard "no training on competitor-API outputs" rule into eng + legal process. (This makes the ROADMAP's "Gemini as offline teacher only" precise:

Comparable products: [Veo](https://www.veo.com/) · [Veo Help Center](https://support.veo.com/hc/en-us/articles/4459159526929-Overview-of-cam-veo-co-Record-manage-and-upload-footage-with-your-Veo-Cam-1) · [Latterly: Hudl competitors](https://www.latterly.org/hudl-competitors/) · [Apple: Behind the Design ? SwingVision](https://developer.apple.com/news/?id=0pg4dthn) · [SwingVision FAQ](https://swing.vision/faq) · [Sportsvisio: coach video-analysis comparison](https://www.sportsvisio.com/stories/video-analysis-software-for-coaches-complete-comparison)

Training backends & harness (\$5A): [karpathy/nanochat (harness reference)](https://github.com/karpathy/nanochat) · [HF nanoVLM (minimal VLM)](https://github.com/huggingface/nanoVLM) · [nanoVLM blog](https://huggingface.co/blog/nanovlm) · [Tinker (GA + vision)](https://thinkingmachines.ai/blog/tinker-general-availability/) · [Tinker model lineup](https://tinker-docs.thinkingmachines.ai/model-lineup) · [Tinker ToS (weights/no-train)](https://thinkingmachines.ai/legal/tos/) · [Fireworks VLM fine-

tuning](https://fireworks.ai/blog/vlm-tuning) · [Together AI fine-tuning](https://www.together.ai/fine-tuning) · [Predibase: fine-tune & serve VLMs](https://predibase.com/blog/how-to-fine-tune-and-serve-vlms-in-predibase) · [Predibase RFT/GRPO](https://docs.predibase.com/user-guide/fine-tuning/grpo) · [ms-swift (native video/audio)](https://github.com/modelscope/ms-swift) · [LLaMA-Factory (VLMs)](https://github.com/hiyouga/LlamaFactory) · [Unsloth (Qwen3-VL)](https://qwen.readthedocs.io/en/latest/training/unsloth.html) · [Fine-tuning tools 2026 (LoRA/QLoRA)](https://www.index.dev/blog/top-ai-fine-tuning-tools-lora-vs-qlora-vs-full)

****Owned model & licensing (\$5A):**** [Qwen2.5-VL](https://qwenlm.github.io/blog/qwen2.5-vl/) · [Qwen2.5 license split](https://qwenlm.github.io/blog/qwen2.5/) · [Qwen3-VL](https://github.com/QwenLM/Qwen3-VL) · [Gemma 3 (multimodal)](https://huggingface.co/blog/gemma3) · [Gemma Terms of Use](https://ai.google.dev/gemma/terms) · [Gemma license-risk analysis](https://wcr.legal/google-gemma-license-risks/) · [Gemma 4 / Apache-2.0](https://www.mindstudio.ai/blog/what-is-gemma-4-apache-2-license-commercial-ai-deployment) · [Llama 3.2 license (700M MAU + naming)](https://www.llama.com/llama3_2/license/) · [gpt-oss (text-only, Apache-2.0)](https://openai.com/index/introducing-gpt-oss/) · [Gemini API terms (compete clause)](https://ai.google.dev/gemini-api/terms) · [OpenAI terms](https://openai.com/policies/row-terms-of-use/) · [Anthropic: outputs & training](https://support.claude.com/en/articles/12326764-can-i-use-my-outputs-to-train-an-ai-model) · [QLoRA practical](https://apxml.com/courses/fine-tuning-adapting-large-language-models/chapter-4-parameter-efficient-fine-tuning/practice-qlora-fine-tuning) · [VLM LoRA cookbook](https://huggingface.co/learn/cookbook/en/fine_tuning_vlmtrl) · [GDPR Art. 9](https://gdpr-info.eu/art-9-gdpr/) · [athletes & biometric data](https://www.architectureandgovernance.com/applications-technology/biometric-data-and-athletes-privacy-law-and-compliance-implications/) · [federated learning + DP](https://research.google/blog/federated-learning-with-formal-differential-privacy-guarantees/) · [self-hosted LLM VRAM/cost](https://www.gpu-mart.com/guides/self-hosted-llm) · [Gemini API pricing](https://ai.google.dev/gemini-api/docs/pricing)

See also

- COMMERCIALIZATION.md ? C12 (SaaS foundations), C4 (paid-LLM), economics.

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Next steps (live) ? owned-model + multisport

****Updated:**** 2026-06-09 (end of the quality-iteration + multisport-recon session).
****Authoritative roadmap:****
 `docs/OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md`. ****Latest blow-by-blow:**** memory `current-state.md` / `session-handoff.md`.
 This file = the prioritized "what to do next," refreshed each session.

The recommendation (one paragraph)

Build the ****quality-first extraction harness**** before anything else: a learned ****TAL head**** (ActionFormer/ASFormer, MIT)
 over trajectory features is the robust path ? it ****decouples from the multi-court "background games" confound**** that the
 unsupervised heuristic (and audio, and spatial court-gating) can't escape (all three tested + failed; it's a ***temporal***
 rally-vs-non-rally problem). ****Framework = ms-swift, base = Qwen3-VL**** (Gemma-4 AMBER/bench)
 come ****later**** for the
 conversational/grounding upgrade ? start with the head. ****On-device is deferred**** (export-clean ops now so it stays
 reachable). Everything stays ****MIT/Apache-2.0/BSD**** (code + data + weights). Bottleneck = ****data + per-sport trajectory detectors****, not model architecture.

Latest findings (2026-06-09)

- ****6-match badminton LOVO:**** learned prototype ****0.583**** vs heuristic ****0.480**** (learned leads since n=3, AUC 0.83?0.92,
 decoupled from multi-court). The "contrast" metric (in-rally-dead-time visibility) predicts heuristic F1 monotonically.

- **?? Multisport (Roora pickleball + TT):** labels **ingest cleanly** (`curate convert-cvat` ? 6 rallies/sport, sane).

WASB-transfer SURPRISE (directional, n=6 each ? TINY): the badminton shuttle detector segments **TABLE TENNIS** at

F1 0.909 out-of-the-box (!) but **pickleball only 0.308**. ? **The per-sport-detector blocker is SPORT-DEPENDENT:** TT may

be servable with WASB short-term; pickleball needs a proper (MIT/Apache) ball detector. Confirm/deny with more videos.

- **Owner field notes (2026-06-09):** camera height/distance **varied on purpose** (great generalization data); **multi-court adjacent-court SOUND dominates the signal** ? empirically reconfirms audio-foreground isolation won't work + the

robustness-first learned-model path. Multi-court is the realistic (academy) environment, so this **is** the target distribution.

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757842c 2026-06-08 docs(platform): address review ? v1-vs-vision scope, Hatchet framing, BYO ops burden

3fd7232 2026-06-08 docs: handoff-consistency sweep + strategy checkpoint + CLAUDE.md (#59)

8fe780e 2026-06-08 docs(commercialization): group dashboard Pending -> In-flight -> Completed + add Priority (P0/P1/P2) column

4d04a73 2026-06-09 Quality iteration: human-GT multi-video LOVO + learned prototype + multi-court finding + owned-model study/plan (#61)

8. user (2026-06-10T05:43:12.054Z)

Structured output provided successfully

9. assistant (2026-06-10T05:43:15.942Z)

Verification complete.